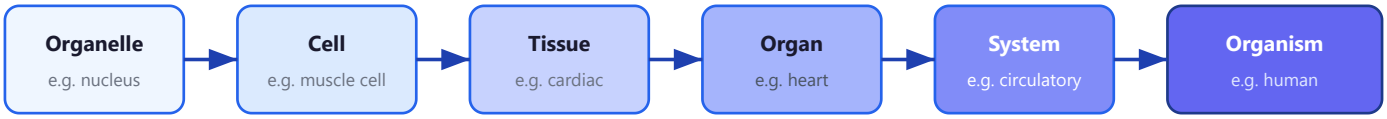
RAG Learning

2.1 Levels of Organisation & Cell Structure

Edexcel International GCSE Biology (4BI1) — Paper 1B/1BR

2.1.1 The Hierarchy of Biological Organisation

Living organisms are built from smaller units combined to form larger, more complex structures: **organelles** → **cells** → **tissues** → **organs** → **organ systems** → **organism**. Each level adds function and integration that the level below cannot achieve alone.



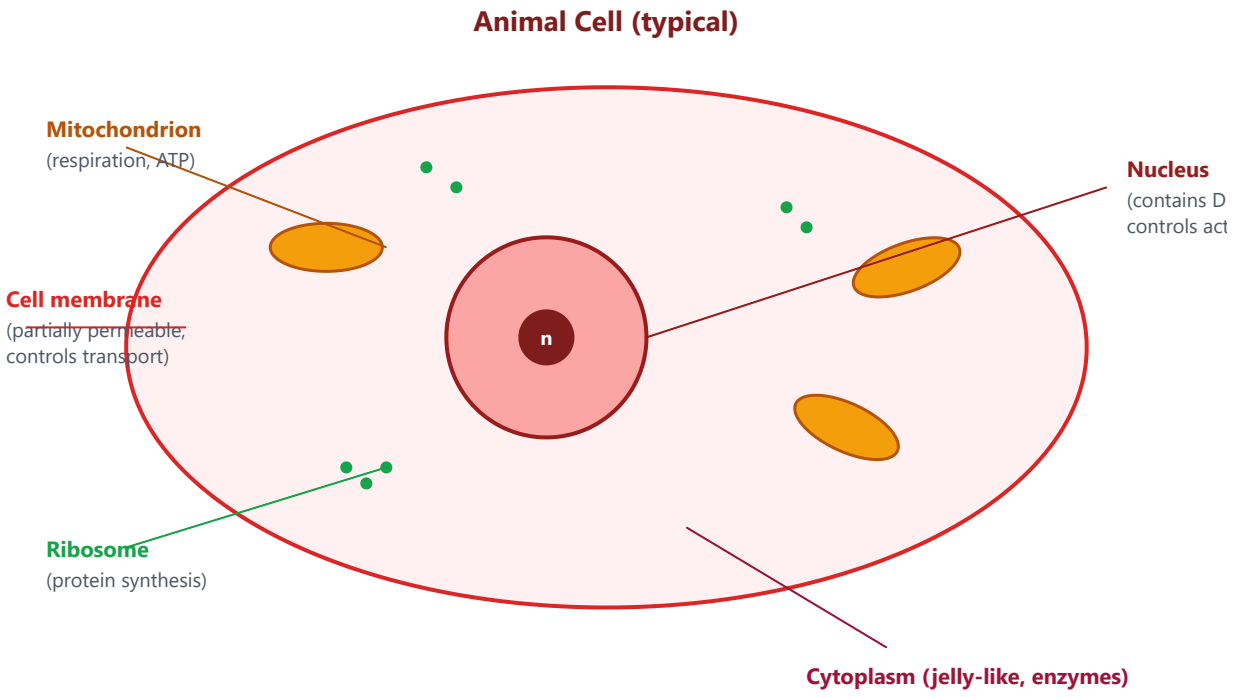
Increasing scale and complexity →

Level	Definition	Example
Organelle	A specialised structure inside a cell with a specific function.	Mitochondrion (site of respiration); nucleus (contains DNA).
Cell	The basic unit of life — the smallest structure capable of carrying out MRS GREN alone.	Red blood cell; palisade mesophyll cell.
Tissue	A group of similar cells working together to perform a specific function.	Muscle tissue contracts; xylem tissue transports water.
Organ	A group of different tissues working together to perform a broader function.	Heart (cardiac muscle + valves + connective tissue); leaf.

© RAG Learning · Not for redistribution

Organ system	Several organs that work together to carry out a major process.	Digestive system (stomach, intestine, liver, pancreas).
Organism	A whole, individual living thing.	A person; an oak tree; a bacterium.

2.1.2 The Animal Cell

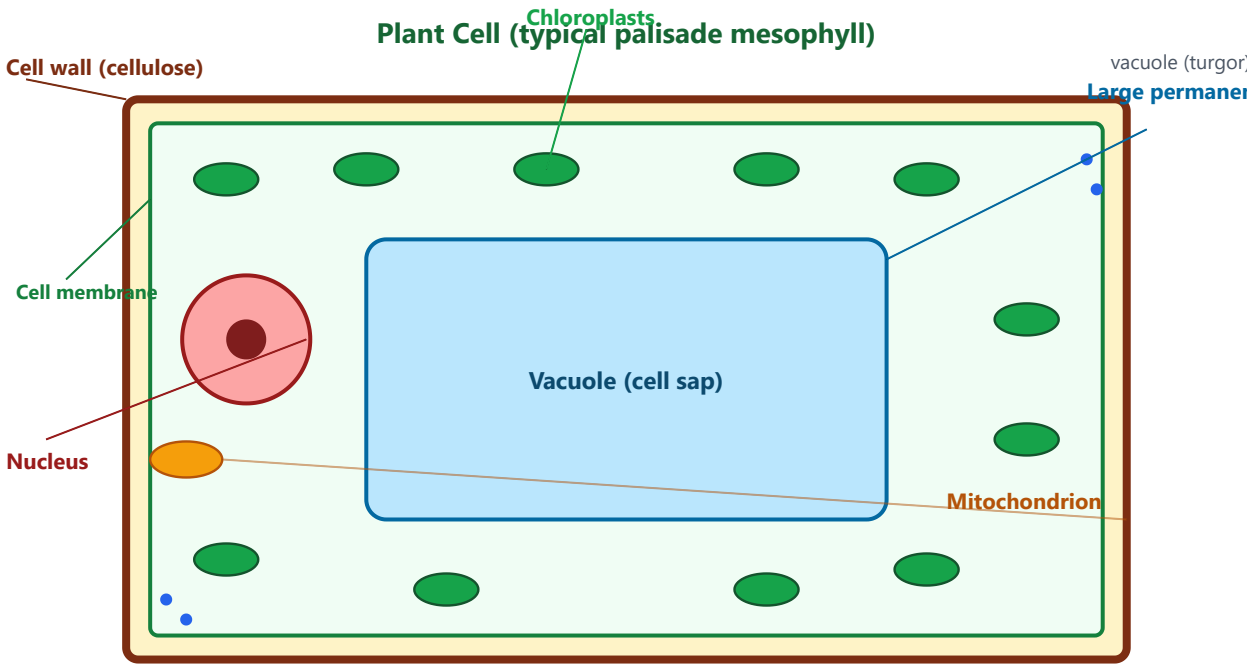


Functions of animal-cell components

Structure	Function
Nucleus	Contains the genetic material (chromosomes of DNA) and controls all cell activities. The nucleolus inside makes ribosomes.
Cytoplasm	Jelly-like fluid where most chemical reactions happen (e.g. glycolysis, protein synthesis). Holds organelles in place.
Cell membrane	Phospholipid bilayer with proteins. Partially permeable — controls which substances enter and leave the cell.

Mitochondrion	Site of aerobic respiration. Glucose + oxygen are converted to CO_2, water and ATP. Cells with high energy demands (e.g. muscle, liver) contain many mitochondria.
Ribosomes	Tiny structures where proteins are made by joining amino acids in the sequence coded by mRNA. Found free in cytoplasm and on rough endoplasmic reticulum.

2.1.3 The Plant Cell



Additional plant-only components

Structure	Material	Function
Cell wall	Cellulose	Strong and fully permeable. Supports the cell, prevents bursting when water enters, gives the plant rigidity.
Chloroplasts	Membrane-bound, contain chlorophyll (pigment)	Site of photosynthesis — absorb light energy to make glucose from CO_2 and water. Only present in green, photosynthesising parts of the plant.

Large permanent vacuole	Membrane-bound sac of cell sap (water + dissolved substances)	Keeps the cell **turgid** so the plant is upright. Also stores dissolved sugars, pigments, waste and ions.

Exam Tip: A frequent Edexcel question is: "*State three structures found in a plant cell that are not found in an animal cell.*" Always list **cellulose cell wall**, **chloroplasts**, and **large permanent vacuole**. Never just "cell wall" — bacteria have a cell wall too but a different one.

2.1.4 Specialised Cells

Multicellular organisms have many cells that are **specialised** — their shape and contents are adapted to a specific function. Here are the key examples you must know for 4BI1.

Cell	Function	Adaptations
Red blood cell	Transport oxygen from lungs to tissues	Biconcave disc — large surface area for gas exchange; no nucleus — more room for haemoglobin; small and flexible — fits through capillaries.
Sperm cell	Fertilise an ovum	Streamlined head + long tail (flagellum) for swimming; many mitochondria in mid-piece for ATP; acrosome with enzymes to penetrate egg; haploid nucleus.
Root hair cell	Absorb water and mineral ions from soil	Long narrow extension — large surface area; thin wall — short diffusion distance; many mitochondria — ATP for active transport of ions.
Palisade mesophyll cell	Photosynthesis in leaf	Many chloroplasts concentrated at the top; long column shape packed together beneath upper epidermis — maximises light absorption.
Ciliated epithelial cell	Sweep mucus up and out of the trachea	Cilia (hair-like projections) that beat rhythmically; goblet cells nearby produce mucus that traps dust and microbes.

© RAG Learning · Not for redistribution

Nerve cell (neurone)	Transmit electrical impulses	Long axon — rapid signal over distance; myelin sheath — insulates and speeds impulse; dendrites — multiple connections to other neurones.
Muscle cell	Contract to produce movement	Packed with actin and myosin filaments that slide past each other; many mitochondria for ATP.

2.1.5 The Microscope and Magnification

Most cells are too small to see with the naked eye. A **light microscope** uses glass lenses and visible light to magnify an image up to ×1500. Under favourable conditions it resolves features about 200 nm apart — good enough to see most cells and nuclei but not individual organelles in detail. An **electron microscope** uses a beam of electrons instead of light and can resolve features as small as 0.1 nm, revealing ribosomes, cristae in mitochondria, and even DNA.

Magnification formula

> **Magnification = image size ÷ actual size**
> The formula has no units — it is a ratio. Make sure both measurements are in the **same units** before dividing (convert mm ↔ µm where needed).

Useful unit conversions:

- 1 mm = 1 000 µm = 1 000 000 nm
- 1 µm = 1 000 nm

> **Worked Example:** An image of a plant cell measures **30 mm** across. The real cell is **75 µm** across. Calculate the magnification.
>
> **Step 1 — Convert units:** 30 mm = 30 000 µm.
> **Step 2 — Apply formula:** magnification = 30 000 ÷ 75 = **×400**.

Worked Example: Under a ×1000 microscope, a bacterium appears **4 mm** long. What is its actual length?

Rearrange: actual = image ÷ magnification = 4 mm ÷ 1000 = 0.004 mm = **4 µm**.

Resolution vs magnification

- **Magnification:** how many times larger the image appears compared with the object. You can increase magnification indefinitely, but the image eventually becomes blurry ("empty magnification").
- **Resolution:** the smallest distance between two points that can be seen as separate. Fixed by the wavelength of the illuminating beam — light microscopes cannot resolve features smaller than ~200 nm.

2.1.6 Preparing a Microscope Slide — Required Practical Skills

To view plant cells (e.g. onion epidermis) under a light microscope:

1. Peel off a **thin layer** of epidermis from the inner side of an onion scale using forceps.
2. Place the tissue flat on a clean microscope slide in a drop of water.
3. Add a drop of **iodine solution** — this stains the nucleus and starch grains dark, making them visible.
4. Lower a coverslip at a 45° angle using a mounted needle to avoid trapping air bubbles.
5. Blot excess fluid. View on low power first, then higher power once the specimen is centred.

Exam Tip: Why lower the coverslip at 45°? **To push out air bubbles**, which refract light and obscure the specimen. Always focus on low power (× 4 or × 10 objective) first to locate the cell, then switch to higher power.

2.1.7 Organelles Compared Across Kingdoms

Structure	Animal	Plant	Bacterium
Nucleus (membrane-bound)	✓	✓	✗ (DNA loop only)
Cytoplasm	✓	✓	✓
Cell membrane	✓	✓	✓
Cell wall	✗	✓ (cellulose)	✓ (peptidoglycan)
Mitochondria	✓	✓	✗
Chloroplasts	✗	✓ (in green tissue)	✗
Large permanent vacuole	✗ (small, temporary only)	✓	✗
Ribosomes	✓	✓	✓ (smaller)
Plasmids	✗	✗	✓

2.1.8 Why Multicellularity and Specialisation Matter

Single-celled organisms rely on a single cell for all life processes — their size is limited by diffusion. Multicellular organisms overcome this through:

- **Specialisation:** different cells efficient at different tasks (e.g. red blood cell for gas transport; ciliated cell for moving mucus).
- **Organisation:** cells → tissues → organs → systems, coordinated by the nervous and hormonal systems.
- **Mass transport:** circulatory and transport systems move substances over distances too great for diffusion alone.
- **Homeostasis:** internal conditions are controlled, allowing much greater complexity and body size than single cells could support.

© RAG Learning · Not for redistribution

Quick recall check: Can you label a plant and an animal cell, list three structures present in a plant cell but absent in an animal cell, calculate magnification, and describe two specialised cells and their adaptations? If so, you've mastered 2.1. Now test yourself with the diagnostic.

© RAG Learning · Not for redistribution